SUPPLEMENTAL MATERIAL

Data S1.

Search Strategy

Embase.com

('stable coronary artery disease'/exp OR ('coronary artery disease'/exp AND stable:ti,ab,kw) OR ('coronary artery disease'/de AND stable:ti,ab,kw) OR (stable NEAR/3 (‘coronary disease*’ OR ‘coronary artery disease*’ OR ‘coronary atherosclerosis’ OR ‘coronary heart disease*’)))

AND

(('percutaneous coronary intervention'/exp OR 'coronary artery bypass graft'/exp) OR (‘percutaneous coronary intervention*’ OR ‘percutaneous coronary angioplast*’ OR ‘balloon angioplasty’ OR ‘coronary angioplasty’ OR ‘PCI’ OR ‘percutaneous coronary revascularization*’ OR ‘coronary artery bypass*’ OR ‘coronary artery bypass graft*’ OR ‘coronary bypass graft*’ OR ‘coronary vein bypass*’ OR ‘coronary venous bypass graft*’ OR ‘aortocoronary bypass*’ OR ‘aortocoronary anastomosis’ OR ‘CABG’):ti,ab,kw)

AND

(drug therapy'/de OR (‘medical therap*’ OR ‘medical-therap*’ OR ‘drug therapy’ OR ‘non surgical’ OR ‘diet’ OR ‘exercise’ OR ‘lifestyle intervention’):ti,ab,kw)

AND

('clinical trial'/de OR 'randomized controlled trial'/de OR 'randomization'/de OR 'single blind procedure'/de OR 'double blind procedure'/de OR 'crossover procedure'/de OR 'placebo'/de OR 'prospective study'/de OR 'randomi?ed controlled' NEXT/1 trial* OR rct OR 'randomly allocated' OR 'allocated randomly' OR ‘randomized trial’ OR 'random allocation' OR allocated NEAR/2 random OR single NEXT/1 blind* OR double NEXT/1 blind* OR (treble OR triple) NEAR/1 blind* OR placebo*)

Ovid-Medline

((Coronary Artery Disease/ AND stable.ti,ab.) OR (Coronary Disease/ AND stable.ti,ab.) OR (stable.mp. ADJ3 ("coronary disease*".mp. OR "coronary artery disease*".mp. OR "coronary atherosclerosis".mp. OR "coronary heart disease*".mp.)))

AND

((exp Percutaneous Coronary Intervention/ OR exp Coronary Artery Bypass/) OR ("percutaneous coronary intervention*" OR "percutaneous coronary angioplast*" OR "balloon angioplasty" OR "coronary angioplasty" OR "PCI" OR "percutaneous coronary revascularization*" OR "coronary artery bypass*" OR "coronary artery bypass graft*" OR

"coronary bypass graft*" OR "coronary vein bypass*" OR "coronary venous bypass graft*" OR "aortocoronary bypass*" OR "aortocoronary anastomosis" OR "CABG").ti,ab.)

AND

Combined Modality Therapy/ OR Drug Therapy/ OR ("medical therap*" OR "medical-therap*" OR "drug therapy" OR "non surgical" OR diet OR exercise OR "lifestyle intervention").ti,ab.)

AND

(randomized controlled trial.pt. OR controlled clinical trial.pt. OR randomized.ab. OR placebo.ab. OR drug therapy.fs. OR randomly.ab. OR trial.ab. OR groups.ab.)

Scopus

(TITLE-ABS-KEY(stable W/3 “coronary disease*”) OR TITLE-ABS-KEY(stable W/3 “coronary artery disease*”) OR TITLE-ABS-KEY(stable W/3 “coronary atherosclerosis”) OR TITLE-ABS-KEY(stable W/3 “coronary heart disease”) OR TITLE-ABS-KEY(stable W/3 “coronary disease*”))

AND

TITLE-ABS-KEY("percutaneous coronary intervention*" OR "percutaneous coronary angioplast*" OR "balloon angioplasty" OR "coronary angioplasty" OR "PCI" OR "percutaneous coronary revascularization*" OR "coronary artery bypass*" OR "coronary artery bypass graft*" OR "coronary bypass graft*" OR "coronary vein bypass*" OR "coronary venous bypass graft*" OR "aortocoronary bypass*" OR "aortocoronary anastomosis" OR "CABG")

AND

TITLE-ABS-KEY("medical therap*" OR "medical-therap*" OR "drug therapy" OR "non surgical" OR diet OR exercise OR "lifestyle intervention")

AND

(INDEXTERMS ("clinical trials" OR "clinical trials as a topic" OR "randomized controlled trial" OR "Randomized Controlled Trials as Topic" OR "controlled clinical trial" OR "Controlled Clinical Trials" OR "random allocation" OR "Double-Blind Method" OR "Single-Blind Method" OR "Cross-Over Studies" OR "Placebos" OR "multicenter study" OR "double blind procedure" OR "single blind procedure" OR "crossover procedure" OR "clinical trial" OR "controlled study" OR "randomization" OR "placebo")) OR (TITLE-ABS-KEY (("clinical trials" OR "clinical trials as a topic" OR "randomized controlled trial" OR "Randomized Controlled Trials as Topic" OR "controlled clinical trial" OR "Controlled Clinical Trials as Topic" OR "random allocation" OR "randomly allocated" OR "allocated randomly" OR "Double-Blind Method" OR "Single-Blind Method" OR "Cross-Over Studies" OR "Placebos" OR "cross-over trial" OR "single blind" OR "double blind" OR "factorial design" OR "factorial trial"))) OR (TITLE-ABS (clinical trial* OR trial* OR rct* OR random* OR blind*))

Cochrane Library

(((stable NEAR/2 “coronary disease*”) OR (stable NEAR/2 “coronary artery disease*”) OR (stable NEAR/2 “coronary atherosclerosis”) OR (stable NEAR/2 “coronary heart disease”))):ti,ab,kw

AND

(("percutaneous coronary intervention*" OR "percutaneous coronary angioplast*" OR "balloon angioplasty" OR "coronary angioplasty" OR "PCI" OR "percutaneous coronary revascularization*" OR "coronary artery bypass*" OR "coronary artery bypass graft*" OR "coronary bypass graft*" OR "coronary vein bypass*" OR "coronary venous bypass graft*" OR "aortocoronary bypass*" OR "aortocoronary anastomosis" OR "CABG")):ti,ab,kw

AND

((“medical therap*” OR “medical-therap*” OR “drug therapy” OR “non surgical” OR diet OR exercise OR “lifestyle intervention”)):ti,ab,kw

Definitions of Nonfatal Myocardial Infarction (MI) in the Included Trials

MASS II

MI was defined as the presence of significant new Q waves in at least two electrocardiographic (ECG) leads or symptoms compatible with MI associated with creatine kinase (CK), MB fraction concentrations that were more than three times the upper limit of the reference range

COURAGE

The prespecified definition of MI (whether periprocedural or spontaneous) required a clinical presentation consistent with an acute coronary syndrome and either new abnormal Q waves in two or more electrocardiographic leads or positive results in cardiac biomarkers. Silent MI, as detected by abnormal Q waves, was confirmed by a core laboratory and was also included as an outcome of MI.

BARI 2D

The MI criteria used were modified from the universal MI definition in that a two-fold elevation of abnormal biomarker profile above the upper limits of normal was used rather than the 99th

percentile. When cardiac troponin and CK-MB were simultaneously acquired, cardiac troponin took precedence over CK-MB in establishing the diagnosis. MI was confirmed if abnormal cardiac biomarkers occurred and there was evidence of angina or angina equivalent symptoms, or ECG or imaging evidence of new myocardial ischemia. Cardiac biomarkers were not routinely collected after coronary revascularization. When they were collected, a 3-fold elevation in CKMB following a PCI procedure and a 10-fold increase in CK-MB following coronary bypass surgery were used as the cut-points to define abnormality.

STICH

A patient must have an increase in cardiac enzymes (CK–MB greater than twice the ULN or troponin T or I greater than three times the ULN) and at least one of the two following criteria:

Typical clinical presentation

OR

Typical ECG changes (evolving ST-segment or T-wave changes in two or more contiguous leads, the development of Q waves in two or more contiguous leads, or the development of new LBBB).

An MI that occurs after a coronary bypass procedure will be adjudicated as a STICH MI only if there are new Q-waves present and the CK-MB is greater than 5 times the ULN and two times the pre – surgery level for CABG.

FAME 2

Within 24 hours after randomization or any PCI:

I. CK-MB above 10 x 99th percentile upper reference limit (URL) determined on a single measurement,

OR

II. CK-MB above 5 x 99th percentile URL determined on a single measurement PLUS at least one of the following:

o new pathological Q waves in at least 2 contiguous leads or new persistent non-rate related left bundle branch block (LBBB),

o angiographically documented native coronary artery occlusion,

o imaging evidence of new loss of viable myocardium

More than 24 hours after randomization:

I. Detection of rise and/or fall of cardiac biomarkers, CK-MB or troponin with at least one value above the 99th percentile of the URL together with evidence of myocardial ischemia with at least one of the following:

o Symptoms of ischemia

o ECG changes indicative of new ischemia (new ST-T changes or new LBBB),

OR

II. Development of pathological Q waves

in ≥ 2 contiguous precordial leads or ≥ 2 adjacent limb leads) of the ECG,

OR

Imaging evidence of loss of viable myocardium or new regional wall motion abnormality

ISCHEMIA

Two versions of MI will be adjudicated in ISCHEMIA: a primary definition and secondary definition. Each definition includes a hierarchy of markers and threshold values as well as a set of rules for diagnosing MI when one or more key elements of the medical record are missing.

The Primary Definition is based upon the Universal Definition of MI, but relies upon site-reported MI decision limits for troponin (which may or may not be the same as the manufacturer 99%URL), and has selected unique marker criteria for MI after PCI or CABG (Type 4a, 5).

The Secondary Definition is also based upon the Universal Definition of Myocardial Infarction, but specifically uses the 99%URL from the assay manufacturer's package insert (which may or may not be the site's MI decision limit) and uses the same supporting criteria (eg. angiographic and ECG) as the UMI definition.

Spontaneous MI

Marker elevation, as outlined below and at least 1 of the following:

- Symptoms of ischemia, usually lasting > 20 minutes in duration
- New ischemic ST and/or T wave and/or Q-wave ECG changes, or new LBBB
- Imaging evidence of new loss of viable myocardium in comparison to the baseline imaging test
- Angiographic evidence of intracoronary thrombus, stent thrombosis (4b) or high-grade in-stent restenosis (≥50%) (4c)

Marker data not available and at least 2 of the following:

- New ischemic ST and/or T wave and/or Q-wave ECG changes, or new LBBB, as described below
- Imaging evidence of new loss of viable myocardium in comparison to the baseline imaging test
- Angiographic evidence of intracoronary thrombus.

Autopsy evidence of a fresh myocardial infarction as stand-alone criterion

Silent MI

This event includes evidence of new silent Q-wave MI detected during routine protocol or clinically obtained ECG follow-up. Silent MI events will be classified as a type 1 MI.

Sudden death MI

MI events in which a presentation consistent with infarction is present but the patient dies before the biomarkers are drawn or within the first few hours of the event before the biomarkers become

positive. Sudden unexpected cardiac death, including cardiac arrest, often with symptoms suggestive of myocardial ischemia, accompanied by presumably new ST-segment elevation, or new LBBB, or evidence of fresh thrombus in a coronary artery by angiography and/or at autopsy, but death occurring before blood samples could be obtained, or at a time before the appearance of cardiac biomarkers in the blood.

PCI-Related MI

CK-MB is the preferred biomarker and takes precedence over troponin. For subjects with normal baseline biomarker level pre-PCI, peri-PCI MI requires a rise in CK-MB to >5-fold the ULN (or a rise in troponin to >35 times the MI Decision Limit/ULN, when CK- MB is unavailable) within 48 hours post-PCI. If pre-PCI cardiac markers (CKMB or cTn) are elevated, they must be stable or falling as indicated by two samples at least 6 h apart. The post-PCI CKMB level should reflect a rise of >20% over pre-PCI levels. In addition to biomarker criteria, peri-PCI MI requires at least one of the following:

- Post- procedure angiographic TIMI 0/1 flow in a major coronary artery or a side branch with reference vessel diameter ≥2.0 mm which had TIMI 2-3 flow at baseline, or TIMI 2 flow in a major coronary artery or a side branch with reference vessel diameter ≥3.0 mm which had TIMI 3 flow at baseline or Type C dissection (NHLBI classification) or greater in the target vessel.

- New ECG changes (ST segment elevation or depression >0.1mV in 2 contiguous leads), new pathologic Q-waves in ≥2 contiguous leads, or new persistent LBBB present on a post-PCI ECG obtained at least 30 minutes and up to 48 hours post procedure in the absence of any intervening coronary event between the time of the PCI procedure and the ECG showing changes.

CABG-Related MI

CK-MB is the preferred serum biomarker and takes precedence over cardiac troponin. For subjects with normal baseline biomarker level pre-CABG, peri-CABG MI requires a rise in CK-MB to >10-fold the ULN (or a rise in troponin to >70 times MI Decision Limit/ULN when CK-MB is unavailable) within 48 hours post-CABG. In addition to biomarker criteria, peri-CABG MI requires at least one of the following:

- A new substantial wall motion abnormality by cardiac imaging (CEC assessed), except new septal and apical abnormalities. The CEC will have latitude in determining whether a new wall motion abnormality is "substantial" in the context of the clinical event.
- New pathologic Q-waves in ≥2 contiguous leads or new persistent LBBB is present on post CABG ECG obtained day 3 post CABG, or hospital discharge, whichever comes earlier in the absence of any intervening coronary event between the time of the CABG procedure and the ECG

ISCHEMIA CKD

Two versions of MI will be adjudicated in ISCHEMIA-CKD: a primary definition and secondary definition. Each definition includes a hierarchy of markers and threshold values as well as a set of rules for diagnosing MI when one or more key elements of the medical record are missing.

The Primary Definition is based upon the Universal Definition of MI, but relies upon site-reported MI decision limits for troponin (which may or may not be the same as the manufacturer 99%URL), and has selected unique marker criteria for MI after PCI or CABG (Type 4a, 5).

The Secondary Definition is also based upon the Universal Definition of Myocardial Infarction, but specifically uses the 99%URL from the assay manufacturer's package insert (which may or may not be the site's MI decision limit) and uses the same supporting criteria (eg. angiographic and ECG) as the UMI definition.

Spontaneous MI (Types 1, 2, 4b, 4c)

Diagnosis of spontaneous MI will be satisfied by a clinical setting consistent with acute myocardial ischemia and any one or more of the following criteria:

Marker elevation, as outlined below and at least 1 of the following:

- Symptoms of ischemia, usually lasting > 20 minutes in duration
- New ischemic ST and/or T wave and/or Q-wave ECG changes, or new LBBB, as described below
- Imaging evidence of new loss of viable myocardium in comparison to the baseline imaging test
- Angiographic evidence of intracoronary thrombus, stent thrombosis (4b) or high- grade in-stent restenosis (≥50%) (4c)

Marker data not available and at least 2 of the following:

- New ischemic ST and/or T wave and/or Q-wave ECG changes, or new LBBB, as described below
- Imaging evidence of new loss of viable myocardium in comparison to the baseline imaging test
- Angiographic evidence of intracoronary thrombus.

Autopsy evidence of a fresh myocardial infarction as stand-alone criterion

Spontaneous MI Marker Criteria

Troponin, including high-sensitivity troponin, is the preferred biomarker and takes precedence over CK-MB for both definitions.

Primary Definition: Preferentially uses a troponin threshold value reported as MI Decision Limit or the Upper Limit of Normal (ULN). Marker elevation is defined as troponin > ULN/MI decision limit. If troponin is not done or not available, then CK-MB > ULN will qualify. If both troponin and CKMB are not done or not available, then CK > 2 x ULN will qualify.

Secondary Definition: Preferentially uses a troponin threshold reported by the manufacturer, namely, the manufacturer 99th percentile. Marker elevation is defined as troponin > 99th percentile. If the troponin 99th percentile is not reported, then troponin > ULN will qualify. If troponin is not done or not available, then CK-MB > ULN will qualify. If both troponin and CK-MB are not done or not available, then CK > 2 x ULN will qualify.

Spontaneous MI ECG Criteria

ECG criterion is considered to be met if any of the following:

ST elevation: New ST elevation at the J-point in two contiguous leads with the cutpoints:

≥ 0.2 mV in men >age 40 and ≥ 0.25mV in men <40 years or ≥ 0.15 mV in women in leads V2–V3 and/or ≥ 0.1 mV in other leads, or new LBBB.

Any new Q-wave in leads V2–V3 ≥ 0.02 seconds or QS complex in leads V2 and V3 or Q-wave ≥ 0.03 seconds and ≥ 0.1 mV deep or QS complex in leads I, II, aVL, aVF, or V4–V6 in any two leads of a contiguous lead grouping (I, aVL, V6; V4–V6; II, III, and aVF) or R-wave ≥ 0.04 seconds in V1–V2 and R/S ≥ 1 with a concordant positive T- wave in the absence of a conduction defect.

ST depression and/or T-wave changes, new horizontal or down-sloping ST depression ≥ 0.05 mV in two contiguous leads; and/or T -wave inversion ≥ 0.1 mV in two contiguous leads. The ST-T wave criteria only apply in the absence of findings that would preclude ECG analysis such as LBBB, LVH with repolarization abnormalities, pre-excitation and pacemakers.

Silent MI

This event includes evidence of new silent Q-wave MI detected during routine protocol or clinically obtained ECG follow-up. Silent MI events will be classified as a type 1 MI.

Sudden death MI (Type 3)

MI events in which a presentation consistent with infarction is present but the patient dies before the biomarkers are drawn or within the first few hours of the event before the biomarkers become positive. Sudden unexpected cardiac death, including cardiac arrest, often with symptoms suggestive of myocardial ischemia, accompanied by presumably new ST-segment elevation, or new LBBB, or evidence of fresh thrombus in a coronary artery by angiography and/or at autopsy, but death occurring before blood samples could be obtained, or at a time before the appearance of cardiac biomarkers in the blood.

PCI-Related MI (Type 4a)

Primary Definition

CK-MB is the preferred biomarker and takes precedence over troponin. For subjects with normal baseline biomarker level pre-PCI, peri-PCI MI requires a rise in CK-MB to >5-fold the ULN (or a rise in troponin to >35 times the MI Decision Limit/ULN, when CK- MB is unavailable) within 48 hours post-PCI. If pre-PCI cardiac markers (CKMB or cTn) are elevated, they must be stable or falling as indicated by two samples at least 6 h apart. The post-PCI CKMB level should reflect a rise of >20% over pre-PCI levels. In addition to biomarker criteria, peri-PCI MI requires at least one of the following:

- Post- procedure angiographic TIMI 0/1 flow in a major coronary artery or a side branch with reference vessel diameter ≥2.0 mm which had TIMI 2-3 flow at baseline, or TIMI 2 flow in a major coronary artery or a side branch with reference vessel diameter ≥3.0 mm which had TIMI 3 flow at baseline or Type C dissection (NHLBI classification) or greater in the target vessel.
- New ECG changes (ST segment elevation or depression >0.1mV in 2 contiguous leads), new pathologic Q-waves in ≥2 contiguous leads, or new persistent LBBB present on a post-PCI ECG obtained at least 30 minutes and up to 48 hours post procedure in the absence of any intervening coronary event between the time of the PCI procedure and the ECG showing changes.

Secondary Definition

Elevation of troponin values >5 X 99th percentile URL within 48 hours post -PCI in patients with normal baseline troponin values pre-PCI AND a rise of troponin values >20% if the

baseline values are elevated pre-PCI and are stable or falling. If the troponin 99th percentile is not available, the MI Decision Limit / ULN may be used. If troponins are not available, CKMB elevation >5 X ULN will be used.

In addition to biomarker criteria, peri-PCI MI requires at least one of the following:

- Symptoms suggestive of myocardial ischemia (≥20 min)
- New ischemic ST changes or new pathological Q waves. (see "ECG Criteria" above) Note
- the UMI definition uses ≥0.05 mV of STD whereas the ISCHEMIA definition uses ≥ 0.1mV
- for PCI related ECG criteria
- Angiographic evidence of a flow limiting complication, such as loss of patency of a side
- branch, persistent slow-flow or no re-flow, embolization, or Type C dissection (NHLBI
- classification) or greater in the target vessel.
- Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.

CABG-Related MI (Type 5)

Primary Definition

CK-MB is the preferred serum biomarker and takes precedence over cardiac troponin. For subjects with normal baseline biomarker level pre-CABG, peri-CABG MI requires a rise in CK-MB to >10-fold the ULN (or a rise in troponin to >70 times MI Decision Limit/ULN when CK-

MB is unavailable) within 48 hrs post-CABG. In addition to biomarker criteria, peri-CABG MI requires at least one of the following:

- A new substantial wall motion abnormality by cardiac imaging (CEC assessed), except new septal and apical abnormalities. The CEC will have latitude in determining whether a new wall motion abnormality is "substantial" in the context of the clinical event.
- New pathologic Q-waves in ≥2 contiguous leads or new persistent LBBB is present on post CABG ECG obtained day 3 post CABG, or hospital discharge, whichever comes earlier in the absence of any intervening coronary event between the time of the CABG procedure and the ECG showing changes.

Secondary Definition

Elevation of troponin values >10 X 99th percentile URL within 48 hrs post -CABG in patients with normal baseline troponin values (≤ 99th percentile URL). If the troponin 99th percentile is not available, the ULN may be used. If troponins are not available, CKMB elevation >10 X ULN will be used. In addition to biomarker criteria, peri-CABG MI requires at least one of the following:

- New pathologic Q waves or new LBBB
- Angiographic evidence of new graft or new native coronary artery occlusion.
- Imaging evidence of new loss of viable myocardium.